RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Computer Science & Engineering

THESIS PROGRESS REPORT

(CSE 4208 – Seminar)
4th Year Even Semester 2024
2020 Series

Tentative Title:

Robust Bengali Sarcasm Detection: Adversarial Fine-Tuning, and Cross-Dataset Evaluation

Submitted By

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1. THESIS IN BRIEF

Item	Description
Course Code	CSE 4208
Course Title	Seminar
Semester	4th Year Even Semester 2024
Thesis Duration	August 2025 – May 2026
Area of Research	Natural Language Processing, Machine Learning, Deep Learning
Research Type	Experimental Research
Tools & Technologies	Python, PyTorch, HuggingFace Transformers, Kaggle (2×T4 GPU), scikit-learn
Dataset Source	Ben-Sarc (25,636), BanglaSarc (5,112), BanglaSarc3 (12,072) — public Bengali corpora

2. ABSTRACT

This thesis explores the detection of Bengali sarcasm by a three-track systematic experiment pipeline built around an explainable, same-dataset comparison with the Q1 Ben-Sarc paper (Lora et al., 2025). Track A constructs the overall thesis pipeline - including dataset preparation, five transformer backbones, adversarial training techniques, regularization literature, ensemble techniques and cross-dataset validation. Track B replicates the ML, deep learning and frozen-encoder transfer families of the reference paper using the same Ben-Sarc dataset. Track C finishes the work by running statistical significance tests, calibration, and producing all publication quality figures.

The best method proposed, BanglaBERT fine-tuned using combined FGM and AWP adversarial training, achieves macro-F1 = 0.8097 on Ben-Sarc, which represents a statistically significant improvement of +0.0650 macro-F1 (+8.7% relative) over the strongest author-side method (LSTM+CNN+PreEmb, macro-F1 = 0.7447). All 22 experimental notebooks, all 5 architecture diagrams, and 12 publication-quality figures are complete.

3. OBJECTIVES

3.1 Main Objectives

1. Achieve the first multi-backbone, multi-technique Bengali sarcasm-detection benchmark on three publicly available datasets.
2. Replicate the three method families of the Q1 reference paper (Lora et al., 2025) on Ben-Sarc to directly and comparatively compare them with the same data set.
3. Use adversarial fine-tuning (FGM + AWP) to show that it achieves better performance compared to frozen-encoder transfer learning in Bengali sarcasm detection.
4. Measure robustness using McNemar significance tests, bootstrap confidence intervals, calibration analysis and cross-dataset transfer evaluation.

3.2 Research Questions

1. Does thesis-side full fine-tuning outperform author-side methods on the same Ben-Sarc dataset under the same evaluation conditions?
2. Are there statistically significant differences between adversarial training and regular BanglaBERT fine-tuning?
3. Is there any reliable transfer between strong in-domain performance to out-of-domain Bengali sarcasm datasets?

4. LITERATURE REVIEW

Papers Reviewed: 10 **Indexed Sources:** Cambridge NLP, NeurIPS, ACL, NAACL, EMNLP, ICML, IEEE

Ref.	Authors	Method	Relevance to This Thesis
[1]	Lora et al. (2025)	Ben-Sarc corpus + ML/DL/Transfer baselines	Introduced primary Bengali sarcasm benchmark; best frozen-encoder at 75.05% accuracy
[2]	Devlin et al. (2019)	BERT: Bidirectional Transformer pre-training	Established modern fine-tuning paradigm for NLP downstream tasks
[3]	Khatun et al. (2023)	BanglaBERT Bengali pre-trained LM	Strongest Bengali-specific transformer; consistent SOTA on Bengali benchmarks
[4]	Conneau et al. (2020)	XLM-RoBERTa cross-lingual representations	Strong multilingual transfer; useful comparison backbone
[5]	Jiang et al. (2020)	SMART / AWP: adversarial weight perturbation	Weight perturbation improves fine-tuning generalization on GLUE/SQuAD
[6]	Barrantes et al. (2020)	FGM/PGD adversarial training survey	Embedding-space perturbation consistently improves NLP robustness
[7]	Liang et al. (2021)	R-Drop: Regularized Dropout	Consistency across dropout samples; marginal gains when adversarial pipeline strong
[8]	Khosla et al. (2020)	Supervised Contrastive Learning	Same-class representations pulled closer; limited benefit for binary Bengali tasks
[9]	Guo et al. (2017)	Calibration of modern neural networks	Strong fine-tuned models tend to be overconfident (high ECE)
[10]	Islam et al. (2022)	Bengali sentiment analysis with transformers	BanglaBERT outperforms mBERT on Bengali sentiment; confirms backbone choice

The main gap in the literature is simple, no previous study has been able to perform full transformer fine-tuning, adversarial training and systematic cross-dataset testing of Bengali sarcasm detection. The best-result in the Lora et al. paper was based on a frozen-encoder paradigm; this thesis specifically examines whether full fine-tuning with adversarial augmentation can outperform this. Also, the thesis reports an actual reproducibility failure, i.e. the original best model that was mentioned in the paper is no longer publicly available, which inspired the reproducibility-first assessment design.

5. METHODOLOGY

5.1 Experimental Design

The thesis is designed in three tracks. The primary thesis pipeline, dataset preparation, fine-tuning of the transformer on five backbones, adversarial training, regularization studies, ensembling, and cross-dataset transfer, is Track A (nb00-14). Track B (nb15 -18) replicates the Lora et al. method families on Ben-Sarc to compare to the same dataset. Track C (nb19 - 22) is used in the testing of significance, calibration, analysis of errors and publication. The entire workflow has been mapped in Figure D4 below.

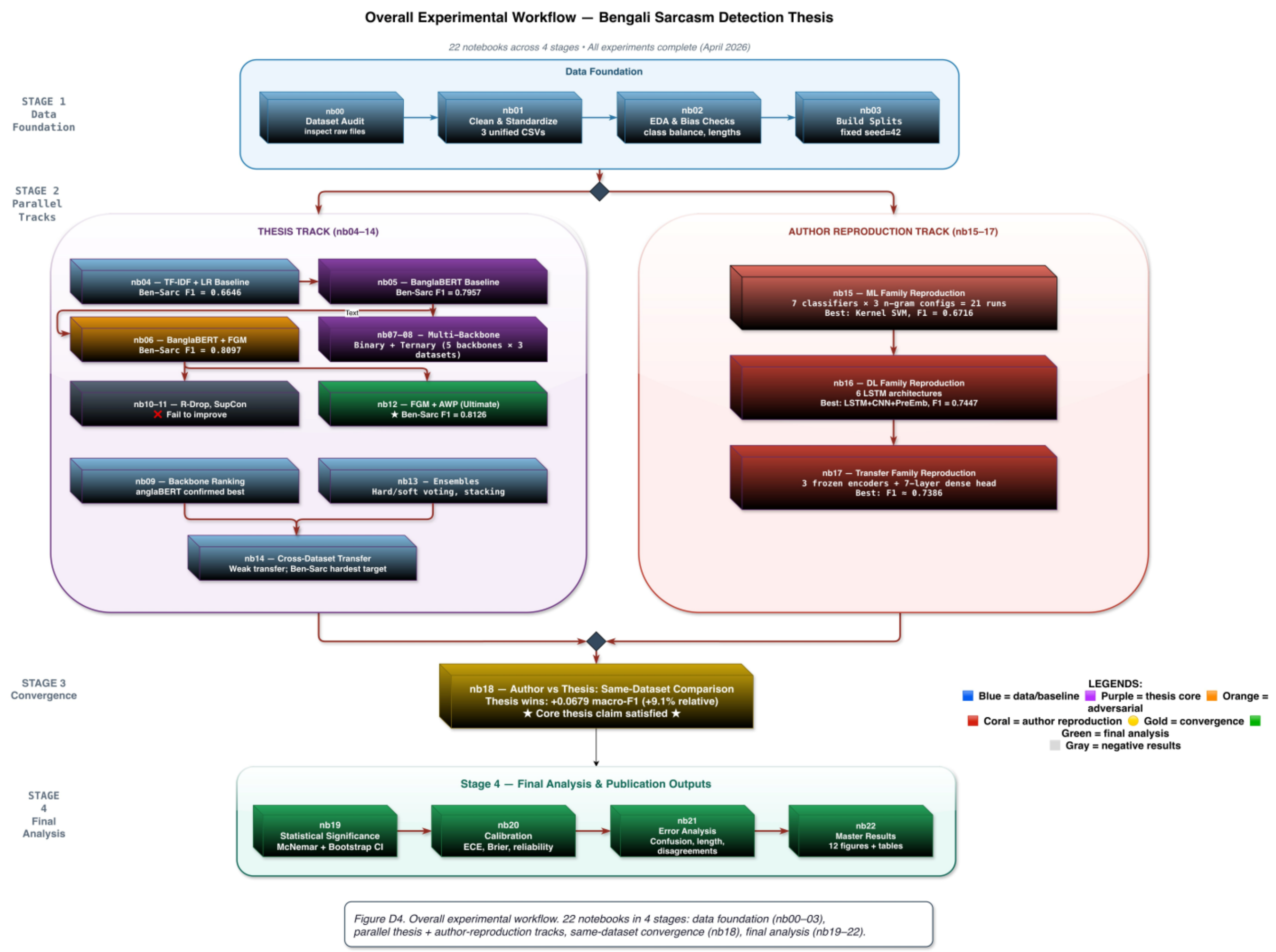

Figure D4. Overall Experimental Workflow (22 Notebooks Across 4 Stages)

Stage 1: Data foundation (nb00–03). Stage 2: Parallel thesis track (nb04–14) and author-reproduction track (nb15–17). Stage 3: Same-dataset convergence (nb18). Stage 4: Final analysis (nb19–22).

5.2 Datasets

Dataset	Rows	Task	Source	Role
Ben-Sarc	25,636	Binary	Facebook	Primary benchmark — same-dataset comparison with Lora et al.
BanglaSarc	5,112	Binary	FB/YT/Blog	Secondary generalization benchmark
BanglaSarc3	12,072	Bin+Tern	Facebook	Ternary ambiguity analysis; binary generalization test

Each dataset was cleaned and standardized to a common schema and divided into fixed train/validation/test sets (80/10/10, stratified, seed=42) in notebooks nb00-nb03. These fixed splits are used identically throughout all the model families, so all reported results can be directly compared.

5.3 Model Families

Family	Methods / Models	Notebook(s)
Classical ML (Thesis)	TF-IDF + LR, Linear SVM, RBF SVM	nb04
Transformer Fine-Tuning	BanglaBERT, MuRIL, XLM-RoBERTa, Sagorsarker BB, mBERT	nb05, nb07–09
Adversarial Training	BanglaBERT+FGM (ε=0.3/0.5/0.7), FGM+AWP, FreeLB (K=3), MuRIL+FGM	nb06, nb12

Family	Methods / Models	Notebook(s)
Regularization / Contrastive	R-Drop, Label Smoothing, SupCon (joint & two-stage)	nb10, nb11
Ensemble	Hard voting, Soft voting, Stacking	nb13
Reproduced (Author-Style)	ML: 7 classifiers × 3 n-gram; DL: 6 LSTM/CNN architectures; Transfer: 3 frozen encoders + 7-layer dense head	nb15–17

5.4 Proposed Method: BanglaBERT + FGM + AWP

The proposed method combines the BanglaBERT backbone (csebuetnlp/banglabert, 110M parameters) with a composite adversarial loop using both FGM and AWP. The training proceeds in five steps per batch: (1) clean forward pass to compute L_clean and populate gradients; (2) AWP perturbs model weights using clean gradients within an ε-ball (adv_eps=10^{-3}); (3) FGM additionally perturbs word embeddings (ε=0.5); (4) adversarial forward pass computes L_adv, gradients accumulate; (5) both perturbations are restored before the optimizer step. The key design choice — both AWP and FGM use clean gradients for perturbation direction — keeps the augmentation well-controlled. Figure D1 illustrates the full architecture.

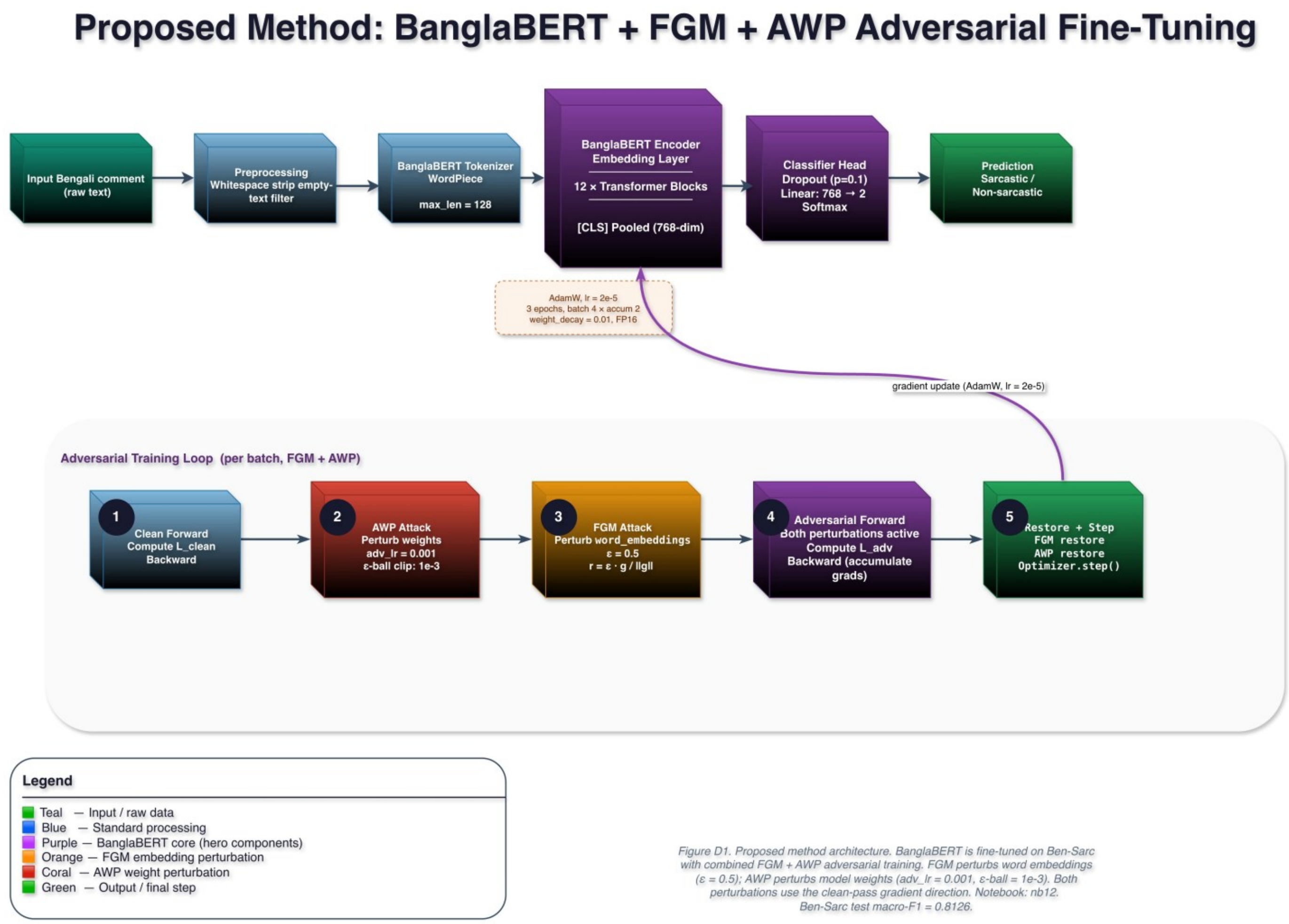

Figure D1. Proposed Method: BanglaBERT + FGM + AWP Adversarial Fine-Tuning

Input → BanglaBERT tokenizer (WordPiece, max_len=128) → 12 Transformer blocks → [CLS] pooled (768-dim) → Classifier head: Dropout(0.1) + Linear(768→2) + Softmax. Adversarial loop: 5-step FGM+AWP per batch (see diagram). Trained 3 epochs, AdamW lr=$2{\times}10^{-5}$, FP16, Kaggle 2×T4 GPU.

6. WORK COMPLETED

All 22 experimental notebooks (nb00–22), all five architecture diagrams (D1–D5), and all twelve publication-quality figures (F1–F12) are complete and frozen.

Completed Task	Status
Dataset auditing, cleaning, EDA, and fixed splits (nb00–03)	DONE
Classical ML baseline: TF-IDF + LR, SVM (nb04)	DONE
BanglaBERT transformer baseline fine-tuning (nb05)	DONE
FGM adversarial training on BanglaBERT (nb06)	DONE
Multi-backbone binary evaluation: 5 backbones × 3 datasets (nb07–09)	DONE
R-Drop, label smoothing, SupCon studies — negative results documented (nb10–11)	DONE
Advanced adversarial sweep: FGM ε-sensitivity, FreeLB, FGM+AWP → macro-F1=0.8097 (nb06/nb12)	DONE
Ensemble methods + cross-dataset transfer (nb13–14)	DONE
Author-style ML, DL, and frozen-transfer reproduction on Ben-Sarc (nb15–17)	DONE
Same-dataset author-vs-thesis comparison (nb18)	DONE
McNemar significance + bootstrap CIs (nb19)	DONE
Calibration: ECE and Brier scores (nb20)	DONE
Error analysis by confusion matrix and text length (nb21)	DONE
All 12 publication-quality figures + summary tables (nb22)	DONE
All 5 architecture diagrams D1–D5 (draw.io)	DONE

7. RESULTS AND DISCUSSION

7.1 Grand Ranking on Ben-Sarc

Figure F1 ranks every evaluated method by macro-F1 on Ben-Sarc binary classification. The thesis-side adversarial methods occupy all top positions (above 0.80 macro-F1), while the best author-side method (LSTM+CNN+PreEmb = 0.7447) sits at rank 10. The visual gap between the two families is the single clearest summary of the thesis's main contribution.

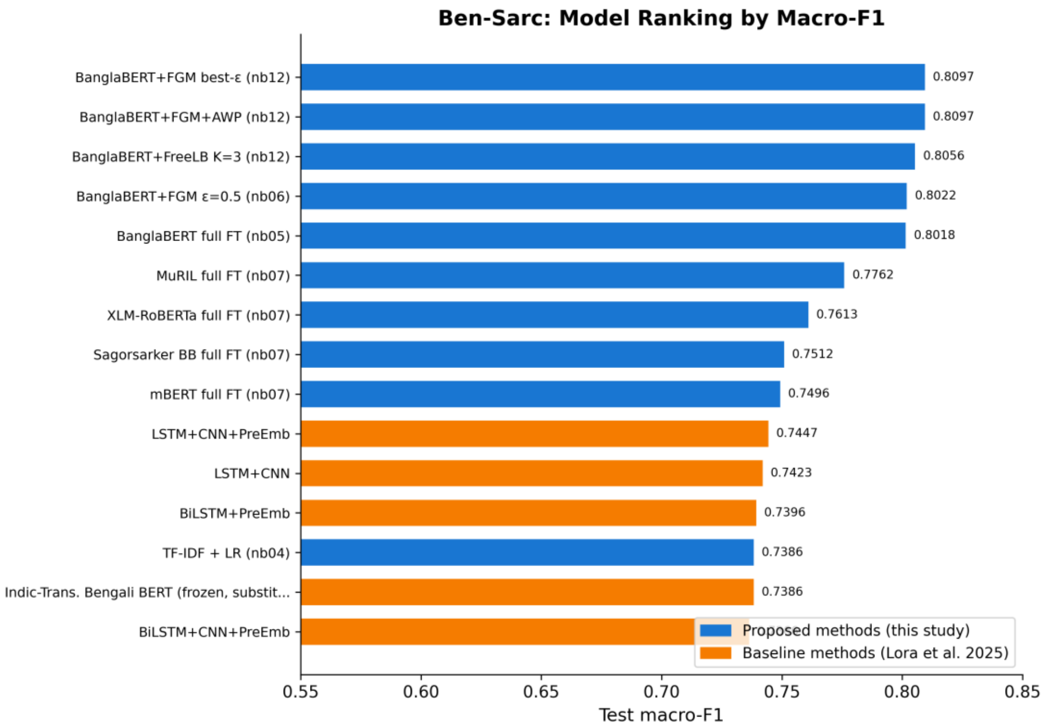

Figure F1. Ben-Sarc Model Ranking by Macro-F1

Blue = proposed thesis methods. Orange = reproduced Lora et al. (2025) methods. Thesis adversarial cluster leads at 0.8022–0.8097; best author-side method reaches 0.7447.

7.2 Head-to-Head: Proposed vs. Lora et al.

Table 7.1 and Figure F10 summarize the central result. BanglaBERT+FGM+AWP outperforms the best reproduced author-side model by 0.0650 macro-F1 (+8.7% relative). The improvement is statistically significant (McNemar $p = 7.98\times10^{-10}$) with non-overlapping 95% bootstrap CIs and P(proposed > reference) = 1.0000 across 1,000 bootstrap resamples.

Table 7.1: Head-to-Head Performance Metrics on Ben-Sarc Test Set

Metric	Proposed (BanglaBERT+FGM+AWP)	Lora et al. (LSTM+CNN+PreEmb)	Delta
Accuracy	80.97%	74.49%	**+6.48%**
Macro Precision	80.97%	74.49%	**+6.48%**
Macro Recall	80.97%	74.57%	**+6.40%**
Macro F1	0.8097	0.7447	**+0.0650 (+8.7%)**

Note: All values are percentages for Accuracy, Precision, and Recall; Macro F1 is reported as a decimal. Delta = Proposed − Lora et al. Both models evaluated on the same Ben-Sarc test split (seed = 42).

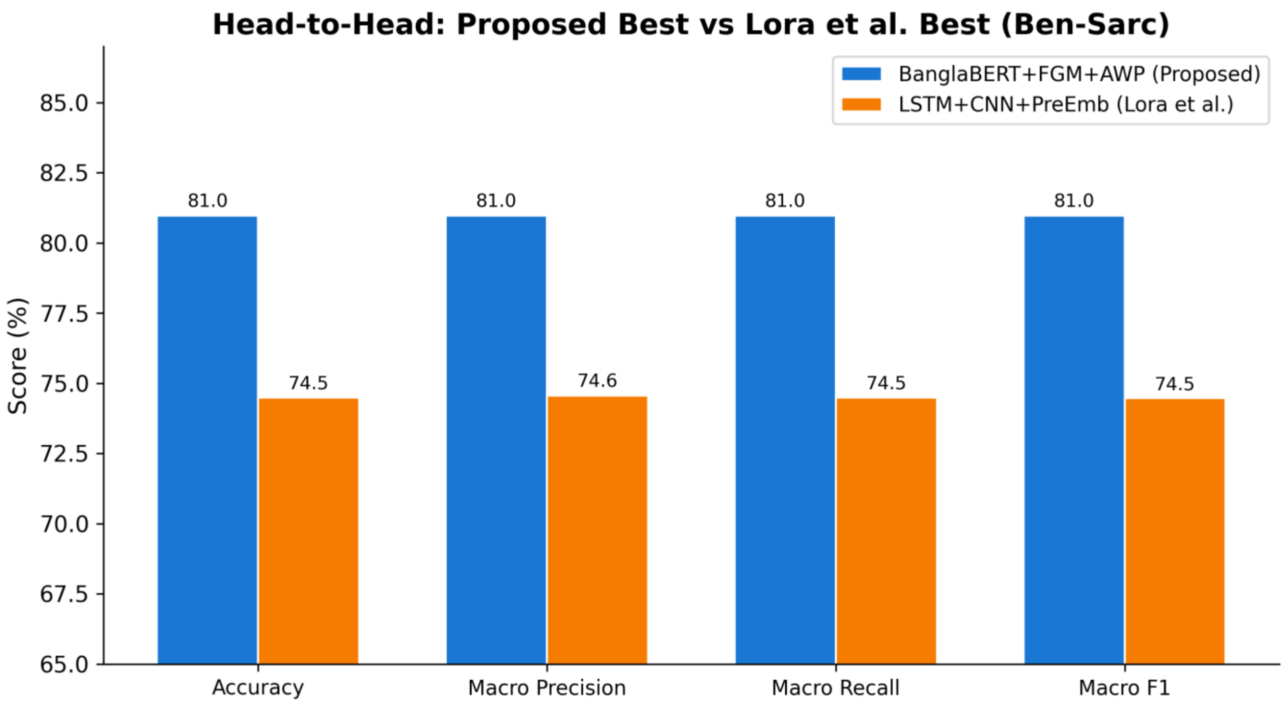

Figure F10. Head-to-Head Metrics: Proposed Best vs. Lora et al. Best (Ben-Sarc)

Proposed BanglaBERT+FGM+AWP (blue) vs. LSTM+CNN+PreEmb reproduced (orange). ~6.5 pp lead across all four metrics.

7.3 Confusion Matrix Analysis

The confusion matrices in Figure F3 show that the proposed model makes 488 total errors (242 FP + 246 FN) versus 654 for the reference model (362 FP + 292 FN) — a 25.4% reduction in absolute error count. The proposed model also shows a more balanced error profile, with nearly equal false positives and false negatives, while the author-side model is biased toward false positives (misclassifying non-sarcastic as sarcastic more often).

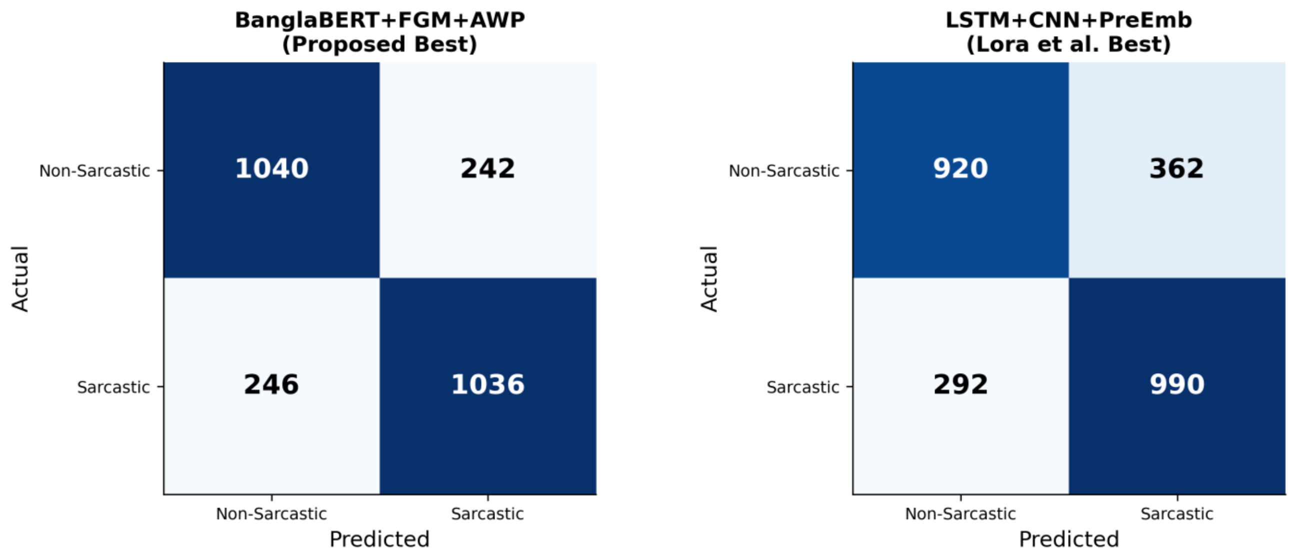

Figure F3. Confusion Matrices: Proposed Best (BanglaBERT+FGM+AWP) vs. Lora et al. Best (LSTM+CNN+PreEmb)

Left: 1040 TP + 1036 TN, 488 total errors. Right: 920 TP + 990 TN, 654 total errors. Proposed model reduces total errors by 25.4%.

7.4 Calibration Analysis

Figure F6 shows an important counterintuitive finding: the proposed BanglaBERT model, despite its superior accuracy, has the highest ECE (0.1294) — meaning it is significantly overconfident. Author-side models are better calibrated despite lower accuracy. This is consistent with the known behavior of fine-tuned transformers and means that probability outputs from BanglaBERT would require post-hoc calibration (e.g., temperature scaling) for deployment use cases requiring reliable confidence estimates.

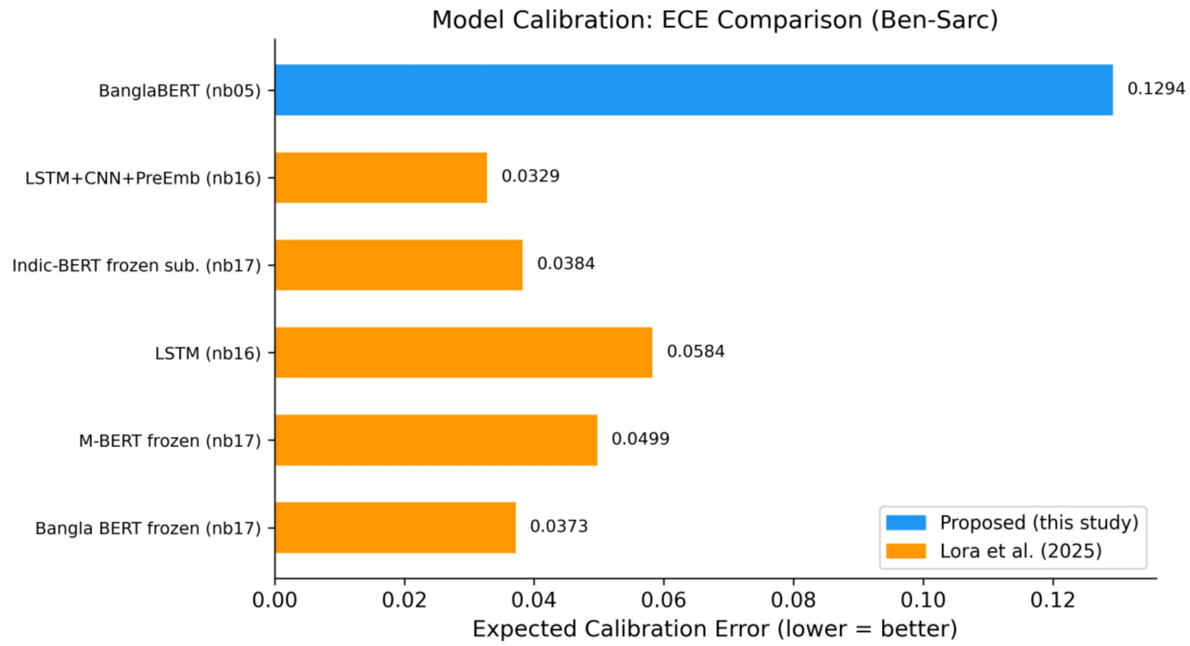

Figure F6. Calibration: Expected Calibration Error Comparison (Ben-Sarc)

7.5 Approach Family Summary

Figure F12 summarizes the results at the approach-family level. The thesis adversarial family (best=0.8097) leads all families; full fine-tuning (best=0.8018) ranks second; and all three author-side families cluster below 0.75. This family-level view is the most accessible high-level summary of the thesis's contributions.

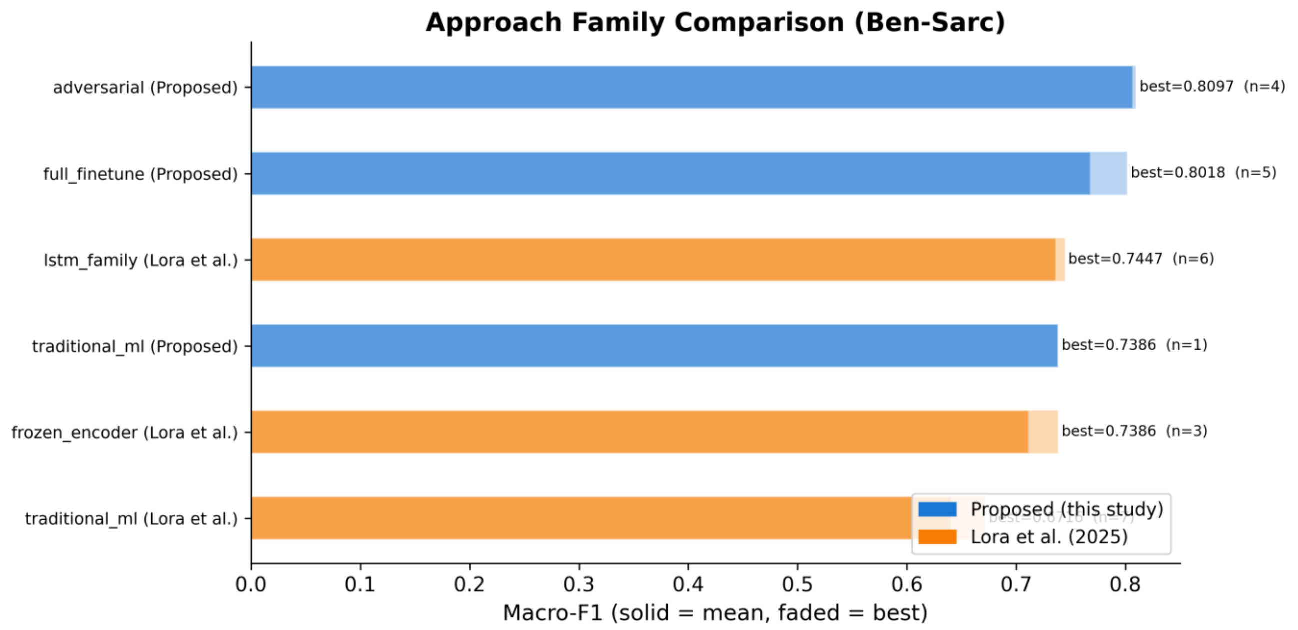

Figure F12. Approach Family Comparison (Ben-Sarc, Macro-F1)

Blue = proposed methods. Orange = Lora et al. Solid = mean family performance; faded = best. Adversarial family leads; all author-side families fall below 0.75.

8. CHALLENGES FACED

1. **Reproducibility gap:** The original Indic-Transformers Bengali BERT referenced in Lora et al. (2025) is no longer publicly accessible on HuggingFace. The thesis documents this explicitly and uses l3cube-pune/bengali-bert as a documented substitute, clearly flagged in all tables and code.
2. **Computational cost of adversarial training:** FGM+AWP loops are roughly 2.5× slower than standard fine-tuning. Managed via FP16 mixed precision on Kaggle 2×T4 GPU, gradient checkpointing, and gradient accumulation (effective batch size 8).
3. **Dataset heterogeneity:** Three datasets with different column names, label formats, and class distributions required a four-notebook standardization pipeline (nb00–03) before any modeling could begin.
4. **Ternary labeling ambiguity:** The ternary task on BanglaSarc3 remained consistently harder across all families. Even the best adversarial model reached only macro-F1 = 0.6618, confirming that the neutral/sarcastic boundary is an open challenge.

9. REMAINING WORK & TIMELINE

Task	Priority	Target Date
Thesis document writing (all chapters)	High	Late April – May 2026
Seminar presentation (PPT) with figures embedded	High	Late April-May 2026
Paper submission	Medium	Post-thesis submission

Timeline

Period	Milestone
Jan 2026	Dataset preparation, EDA, fixed splits (nb00–03)
Feb 2026	Classical ML baseline, transformer fine-tuning, multi-backbone experiments (nb04–09)
Mar 2026	Regularization studies, adversarial sweep, ensemble & cross-dataset (nb10–14)
Apr 2026	Author-method reproduction (nb15–17), comparison (nb18), full analysis (nb19–22), all diagrams

Period	Milestone
Apr–May 2026	Thesis document writing across all chapters + seminar presentation
May 2026	Final submission

10. EXPECTED CONTRIBUTIONS

10.1Academic

1. First multi-backbone, multi-technique Bengali sarcasm detection benchmark with three datasets.
2. Full fine-tuning based on adversarial perturbation significantly outperforms the paradigm of frozen-encoder of the reference paper in terms of relative macro-F1 gain (+8.7%, $p < 0.001$).
3. Evidence of a tangible reproducibility gap in the Ben-Sarc paper, which inspired reproducibility-first practice in Bengali NLP.
4. Publishable negative results: This task cannot be improved by R-Drop, label smoothing or supervised contrastive learning over adversarial fine-tuning.

10.2 Practical

- Open-source pipeline of 22 notebooks: github.com/Sefayet-Alam/Sarcasm_detection
- Architecture diagrams and 12 publication-quality figures ready for journal/conference submission.
- Fixed evaluation protocol reusable for future Bengali NLP benchmarking.

11. REFERENCES

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12. DECLARATION

I hereby declare that this progress report represents my own original work carried out under the supervision of my thesis supervisor at the Department of Computer Science & Engineering, Rajshahi University of Engineering & Technology. All experimental results, figures, and tables presented are generated from my own implemented notebooks and analysis pipelines. Any methods, datasets, or results from prior work have been properly cited.

Signature of Student:	Signature of Supervisor:
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